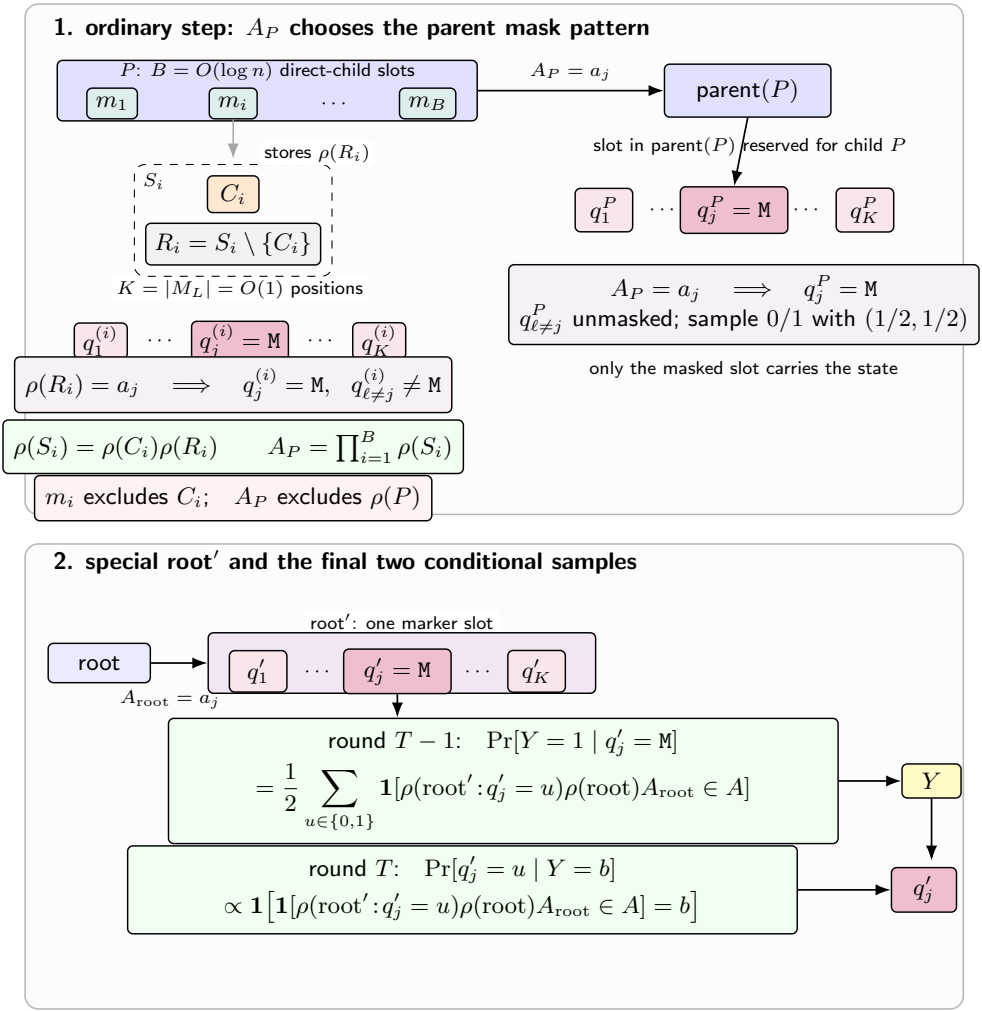
1. ordinary step: $A_P$ chooses the parent mask pattern
$P$: $B = O(\log n)$ direct-child slots
$m_1$
$m_i$
$\cdots$
$m_B$
$A_P = a_j$
parent$(P)$
stores $\rho(R_i)$
$S_i$
$C_i$
$R_i = S_i \setminus \{C_i\}$
$K = |M_L| = O(1)$ positions
slot in parent$(P)$ reserved for child $P$
$q_1^P$ $\cdots$ $q_j^P = \mathtt{M}$ $\cdots$ $q_K^P$
$A_P = a_j \implies q_j^P = \mathtt{M}$
$q_{\ell \neq j}^P$ unmasked; sample $0/1$ with $(1/2, 1/2)$
only the masked slot carries the state
$q_1^{(i)}$ $\cdots$ $q_j^{(i)} = \mathtt{M}$ $\cdots$ $q_K^{(i)}$
$\rho(R_i) = a_j \implies q_j^{(i)} = \mathtt{M}, \quad q_{\ell \neq j}^{(i)} \neq \mathtt{M}$
$\rho(S_i) = \rho(C_i)\rho(R_i) \qquad A_P = \prod_{i=1}^{B} \rho(S_i)$
$m_i$ excludes $C_i$; $A_P$ excludes $\rho(P)$
2. special root′ and the final two conditional samples
root′: one marker slot
root
$q'_1$ $\cdots$ $q'_j = \mathtt{M}$ $\cdots$ $q'_K$
$A_{\mathrm{root}} = a_j$
round $T-1$: $\Pr[Y = 1 \mid q'_j = \mathtt{M}]$
$= \frac{1}{2} \sum_{u \in \{0,1\}} \mathbf{1}[\rho(\mathrm{root}' : q'_j = u)\rho(\mathrm{root})A_{\mathrm{root}} \in A]$
$Y$
round $T$: $\Pr[q'_j = u \mid Y = b]$
$\propto \mathbf{1}\big[\mathbf{1}[\rho(\mathrm{root}' : q'_j = u)\rho(\mathrm{root})A_{\mathrm{root}} \in A] = b\big]$
$q'_j$